SHORT CV OF **AZIM (ABDOLAZIM) REZAEI**

ADDRESS

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EDUCATION

Ph.D. in Computer Science (Aug 2022 – Present), Texas A&M University, Corpus Christi, USA / GPA:3.9
M.Sc. in Software Engineering (2013), University of Malaya, Kuala Lumpur, Malaysia.
B.Sc. in Computer Engineering (2008), Islamic Azad University, Kerman, Iran.

PROFESSIONAL SUMMARY

Research Assistant, Texas A&M University, Corpus Christi, Aug 2022 – Present:

- Author research and technical papers.
- Develop and design innovative research ideas and write research grant proposals.
- Design and implement Transformer, and LLM frameworks.
- Built Graph Neural Network (GNN) and Transformer-based models for multi-modal spatio-temporal Network traffic forecasting.
- Developed LLM-powered frameworks for generating textual descriptions from visual scenes (Vision-to-Text transformation, LangChain etc)
- Integrated Generative AI, RAG and Mixture of Experts (MoE) models into real-time AI systems.
- Engineered time-series features and designed experiments to improve geospatial forecasting accuracy.
- Implemented feedback-driven RL for optimizing privacy-preserving visual systems.

Data Scientist, Dade Pooyan Co, Tehran, Iran, Jun 2016 – Jun 2022:

- Developed and deployed fraud detection models using XGBoost, LightGBM, and deep neural networks.
- Performed data preprocessing, feature engineering, and feature selection on large-scale financial transaction datasets.
- Implemented imbalanced data handling techniques such as SMOTE, up-sampling, and etc.
- Conducted exploratory data analysis (EDA) using Python libraries like Pandas, NumPy, Seaborn, and Matplotlib to identify data patterns and insights.
- Built and maintained real-time analytics dashboards using Power BI to monitor key fraud metrics and system alerts.
- Designed and executed A/B tests and statistical analyses to evaluate model improvements and business impact.
- Wrote technical documentation and reports for model decisions and presented results to stakeholders and executive teams.
- Built and managed data pipelines using SQL and Microsoft SSIS

Data Analytics, University of Applied Sciences, Tehran, Iran, Feb 2014 – Jun 2016:

- Conducted research in graph-based anomaly detection and pattern recognition in online social networks.
- Designed and taught university-level courses in Artificial Intelligence, Data Structures, Algorithms, and Software Engineering.
- Performed data analysis using Python and R to extract meaningful insights from social network datasets.
- Analyzed large datasets to discover behavioral anomalies and community structures in networks.
- Created visual reports and dashboards to present research findings and academic assessments.

Software Developer and Business Analyst, Lateral Co, Kuala Lumpur, Malaysia, May 2011 – Dec 2013

- Conducted stakeholder interviews to elicit business and technical requirements
- Analyzed and documented functional and non-functional requirements
- Created use cases, user stories, and process flow diagrams
- Translated business needs into technical specifications for development teams

- Conducted stakeholder meetings and status updates to gather feedback and align goals
- Worked closely with Product Owners and Scrum Masters to ensure alignment on deliverables
- Implemented test-driven development (TDD) in agile environments
- Designed and optimized Relational Database Management Systems
- Worked in agile/scrum teams

AWARDS & HONORS

- College of Engineering Graduate Scholarship - Spring 2026
- Geospatial Surveying Engineering Scholarship - Spring 2026
- The Best Conference Paper Candidate, The 7th IEEE International Conference on Cognitive Machine Intelligence 2025 (**IEEE CogMI 2025**)
- Selected as Top 11% Papers Invited to **AAAI-26** Three-Minute Presentation Contest
- Scholar Achievement in Graduate Education (SAGE) Fellowship (2025, 2026)
- Scholar Achievement in Graduate Education (SAGE) Fellowship (2024, 2025)
- Scholar Achievement in Graduate Education (SAGE) Fellowship (2023, 2024)
- Scholar Achievement in Graduate Education (SAGE) Fellowship (2022, 2023)
- Best Researcher Award (2020)

SKILLS

Programming & ML Frameworks:
- Python, C++, C#, R
- PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face etc.

AI/ML Techniques:
- Reinforcement Learning (PPO, DQN, etc.)
- Vision-Language Models (CLIP, BLIP, ViLT)
- Generative AI & Large Language Models (LLMs, RAG, MoE)
- Time-Series Forecasting (LSTM, Transformers, GNN)
- Data Mining & Feature Engineering

Cloud & Big Data Tools:
- AWS (S3, Lambda, EC2), Google Cloud, Amazon Bedrock, Amazon Sagemaker
- Docker, Kubernetes, Apache Spark
- SQL, NoSQL, GraphDB

Visualization & Analytics:
- Power BI, Matplotlib, Seaborn

Publications

Rezaei, Abdolazim; Sookhak, Mehdi; " *Parameter Efficient Hybrid Transformer (PEHT) for Network Traffic Prediction via Dynamic Urban Congestion Integration*", IEEE 104th Vehicular Technology Conference: VTC2026-Fall, 2026.

Rezaei, Abdolazim; Sookhak, Mehdi; " *Spatially-Enhanced Quantized Deep Metric Learning in Intelligent Transportation System*", Forty-Third International Conference on Machine Learning (In-progress, targeting **ICML-26**)

Rezaei, Abdolazim; Sookhak, Mehdi; "PEFT-DML: Parameter-Efficient Fine-Tuning Deep Metric Learning for Robust Multi-Modal 3D Object Detection in Autonomous Driving", The 40th Annual Conference on Artificial Intelligence (**AAAI-26**), 2026

Rezaei, Abdolazim; Sookhak, Mehdi; Haghparast, Mahboobeh. "*RL-MoE: An Image-Based Privacy Preserving Approach In Intelligent Transportation System.*" ***IEEE Transactions on Big Data*** (2025).

Taleb, Nabil Abdelaziz Ferhat; **Rezaei, Abdolazim**; Patel, Raj Atulkumar; Sookhak, Mehdi; "*GraphTrafficGPT: Enhancing Traffic Management through Graph-Based AI Agent Coordination*",arXiv preprint arXiv:2507.13511,2025

Rezaei, Abdolazim; Sookhak, Mehdi; Patooghy, Ahmad; "*Privacy-Preserving in Connected and Autonomous Vehicles Through Vision to Text Transformation*",arXiv preprint arXiv:2506.15854,2025

Rezaei, Abdolazim; Yazdinejad, Mohsen; Sookhak, Mehdi; "*Credit Card Fraud Detection Using Tree-Based Algorithms For Highly Imbalanced Data*",2024 IEEE 3rd International Conference on Computing and Machine Intelligence (ICMI),1-6,2024,IEEE

Rezaei, Abdolazim; Kasirun, Zarinah Mohd; Rohani, Vala Ali; Khodadadi, Touraj; "*Anomaly detection in online social networks using structure-based technique*",8th International Conference for Internet Technology and Secured Transactions (ICITST-2013),619-622,2013,IEEE